

Vaibhav Satish

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EDUCATION

University of California, Irvine

Bachelor of Science in Computer Science; GPA: 3.861

– Awards: Dean's Honors List (Every quarter to date)

Irvine, CA

August 2024 – March 2027

WORK EXPERIENCE

Software Development Co-Lead

September 2024 – Present

MedTech@UCI

Irvine, CA

- Co-leading the development of a cross-platform image recognition and memory testing application in collaboration with the UCI SLEEP (Sleep, Learning, and Emotion in Pediatrics) Lab to analyze pediatric cognitive patterns.
- Developed cognitive games using JS and React.js for dementia and Alzheimer's patients' therapeutic use.
- Implemented metrics tracking with React to provide doctors with insights into patient cognitive performance.

Software Development Intern & Instructor

June 2025 – August 2025

Tech Academy of Minnesota

Seattle, WA

- Taught programming to 18 students using Python, HTML, CSS, and JS to build functional web applications.
- Developed hands-on lessons in coding and debugging to strengthen interest in software engineering.
- Adapted complex technical topics for diverse learners, improving technical communication skills.

Artificial Intelligence Researcher

May 2023 – November 2023

Stanford University & UC San Diego

Remote

- Analyzed high-dimensional NFL injury datasets to identify situational risk factors.
- Developed and evaluated predictive models, including Softmax Regression and Neural Networks, using PyTorch and Scikit-Learn.
- Improved prediction accuracy of player injury severity over existing baseline methods through iterative model refinement.
- Documented technical limitations and improvement strategies to guide subsequent research and model iterations.

Artificial Intelligence Researcher

June 2023 – August 2023

Georgia Tech & Cornell University

Remote

- Investigated NLP and computer vision subfields using Python, Scikit-Learn, and TensorFlow.
- Created AI models to detect distracted driving, improving analysis and solution design.
- Presented findings to university researchers, leading to refined project impact and effectiveness.

Software Developer

June 2023 – June 2024

First Robotics, Team 1294: Pack of Parts

Sammamish, WA

- Integrated an autonomous robot alignment system using computer vision for real-time course correction during matches.
- Architected and tested sensor algorithms in Java to improve drivetrain precision and object detection consistency.
- Collaborated with the mechanical engineering sub-team to calibrate software parameters based on physical prototype iterations.
- 1st place at Pacific Northwest District event and qualified for District Championships, demonstrating competitive performance.

PROJECTS

ZOTNest – IrvineHacks 2026

February 2026 – Present

React, Python, FastAPI, Leaflet

- Designed and launched a property searching application based on user research to aid in housing search for 30,000+ UCI students.
- Integrated Leaflet map API for ease of locating the property.
- Integrated ability to sort by price, distance from campus, and based on user needs/wants.
- Scraped Google Maps reviews for each location to provide information about the community.

Driver Delineation – Datathon 2025

April 2025 – April 2025

Python, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib

- Built supervised ML models using PyTorch, NumPy, TensorFlow, and Scikit-Learn to predict driver likelihood from structured data.
- Analyzed structured data to identify key impact variables, translating complex findings into actionable insights for model refinement.
- Trained a neural network with TensorFlow/PyTorch, achieving 90% accuracy on the validation set.

UCI Themed Geoguessr

December 2024 – Present

React, Node.js, TypeScript, C++, Crow API, Vercel

- Led a 4-person team from system architecture to deployment (product lifecycle), prioritizing features based on gameplay importance.
- Implemented custom scoring algorithms and integrated OpenStreetMap APIs for map rendering.
- Developed C++ backend with Crow API for image parsing and accurate gameplay results.

Portfolio Website

May 2025 – September 2025

HTML, CSS, JavaScript, Formspree API, Vercel

- Built a responsive portfolio with asynchronous form handling and client-side validation.
- Integrated Formspree API to facilitate seamless user communication and lead generation.
- Optimized frontend assets and utilized Vercel's global CDN to ensure high performance and sub-second load times across devices.
- Implemented a CI/CD pipeline via GitHub integration for automated deployment and version control management.

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, C#, HTML/CSS

Frameworks: React, Node.js, RESTful APIs, Crow API, OpenStreetMap API, Formspree API, Leaflet, FastAPI

Libraries/Tools: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, Matplotlib, Git, Unity

Core Computer Science: Data Structures, Algorithms, Object Oriented Programming, Client-Server Architecture

Software Development: Agile/Scrum, CI/CD, System Architecture, and Full-stack Development, Product Roadmap, User Stories, Prioritization.

Artificial Intelligence/Machine Learning: Supervised Learning/Unsupervised Learning, NLP, Computer Vision